

3.2

$$\langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

2) $\text{Span}\langle 1, x, x^2 \rangle$ $\vec{v}_1 = 1, \vec{v}_2 = x, \vec{v}_3 = x^2$

1) $\langle 1, 1 \rangle = \int_1^5 dx = [x]_1^5 = 4$

$$\|1\| = \sqrt{\langle 1, 1 \rangle} = 2$$

$$\underline{\underline{\vec{u}_1 = \frac{1}{2} \vec{v}_1 = \frac{1}{2}}}$$

2) $\vec{w}_2 \dots$ orthogonal $\vec{v}_i$ do $\text{Span}\langle \vec{u}_1 \rangle$

$$\vec{w}_2 = \langle \vec{u}_1, \vec{v}_2 \rangle \cdot \vec{u}_1 = \int_1^5 \frac{1}{2} x dx \cdot \frac{1}{2} = \frac{1}{2} \left[\frac{x^2}{2} \right]_1^5 \cdot \frac{1}{2} = \frac{1}{4} \left(\frac{25}{2} - \frac{1}{2} \right) = \frac{24}{8} = 3$$

$$\vec{v}_2 - \vec{w}_2 = x - 3$$

$$\|\vec{v}_2 - \vec{w}_2\| = \|x - 3\| = \sqrt{\langle x - 3, x - 3 \rangle} = \sqrt{\int_1^5 (x - 3)^2 dx} = \sqrt{\int_{-2}^2 t^2 dt} = \sqrt{\left[\frac{t^3}{3} \right]_{-2}^2} = \sqrt{\frac{8}{3} + \frac{8}{3}} = \sqrt{\frac{16}{3}} = \frac{4}{\sqrt{3}}$$

$$x - 3 = t$$

$$x = t + 3$$

$$dx = dt$$

$$x = 1 \rightarrow t = -2$$

$$x = 5 \rightarrow t = 2$$

$$\underline{\underline{\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\|\vec{v}_2 - \vec{w}_2\|} = \frac{\sqrt{3}}{4} (x - 3)}}$$

3) $\vec{w}_3$ orthogonal $\vec{v}_i$ do $\text{Span}\langle \vec{u}_1, \vec{u}_2 \rangle$

$$\vec{w}_3 = \langle \vec{u}_1, \vec{v}_3 \rangle \cdot \vec{u}_1 + \langle \vec{u}_2, \vec{v}_3 \rangle \cdot \vec{u}_2 = \frac{31}{3} + 15,86 \cdot \frac{\sqrt{3}}{4} \cdot (x - 3) = 6x - 7,67$$

$$\int_1^5 \frac{1}{2} x^2 dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_1^5 = \frac{1}{2} \left(\frac{125}{3} - \frac{1}{3} \right) = \frac{124}{6} = \frac{62}{3}$$

$$\int_1^5 \frac{\sqrt{3}}{4} (x - 3) \cdot x^2 = \frac{\sqrt{3}}{4} \int_1^5 x^3 - \frac{3\sqrt{3}}{4} \int_1^5 x^2 = \frac{\sqrt{3}}{4} \cdot \frac{1}{4} \left[x^4 \right]_1^5 - \frac{3\sqrt{3}}{4} \cdot \frac{1}{3} \left[x^3 \right]_1^5 = 13,86$$

$$\vec{v}_3 - \vec{w}_3 = x^3 - 6x + 7,67$$

$$\langle \vec{v}_3 - \vec{w}_3, \vec{v}_3 - \vec{w}_3 \rangle = \int_1^5 (x^3 - 6x + 7,67)^2 dx$$

$$\|\vec{v}_3 - \vec{w}_3\| = 2,33$$

$$\underline{\underline{\vec{u}_3 = \frac{1}{2,33} (x^3 - 6x + 7,67)}}$$

b) Vypočítaj ON báze z ústí a)

$$P_{\perp}(\sin x) = \langle \vec{u}_1, \sin x \rangle \cdot \vec{u}_1 + \langle \vec{u}_2, \sin x \rangle \cdot \vec{u}_2 + \langle \vec{u}_3, \sin x \rangle \cdot \vec{u}_3$$

$$\langle \vec{u}_1, \sin x \rangle \vec{u}_1 = \int_1^5 \frac{1}{2} \sin x \, dx \cdot \frac{1}{2} = 0,064$$

$$\langle \vec{u}_2, \sin x \rangle \vec{u}_2 = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot \sin x \, dx \cdot \frac{\sqrt{3}}{4} (x-3) = -0,65 (x-3)$$

$$\langle \vec{u}_3, \sin x \rangle \vec{u}_3 = \int_1^5 \frac{1}{2,39} (x^2 - 6x + 7,67) \cdot \sin x \, dx \cdot \frac{1}{2,39} (x^2 - 6x + 7,67) = -0,052 (x^2 - 6x + 7,67)$$

$$P_{\perp}(\sin x) = -0,052 x^2 - 0,34 x + 1,62$$

c) $\sqrt{\int_1^5 (\sin x + 0,052 x^2 + 0,34 x - 1,62)^2 \, dx} = 0,32$